Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam E · Form 2: Constraint analysis

Multiple Choice Questions: Select the best answer. (1 point each)

1. A student walks 12 m right from a sensor, then 3 m left in a total of 6 s. Taking right as positive, which pair gives the additional displacement needed to return to the start and the total distance after returning?

(A) -9 m; 24 m

(B) 9 m; 15 m

(C) Both zero

(D) +9 m; 15 m

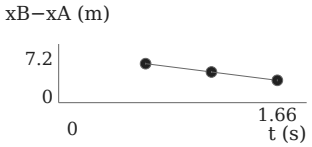
2. A runs at +5 m/s and B at +2 m/s. Record the lead of B ahead of A as a signed separation. Exchange the runners' velocities but leave starting positions unchanged. What is the new intercept minus the original intercept?

(A) 1.89

(B) 0

(C) 0.63

(D) -1.26



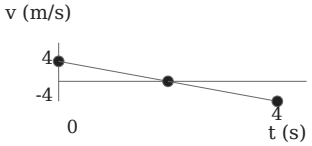
3. A velocity–time graph is a straight line from +4 m/s at t = 0 to -4 m/s at t = 4 s. What is the displacement?

(A) 5 m

(B) -3 m

(C) 0 m

(D) 2 m



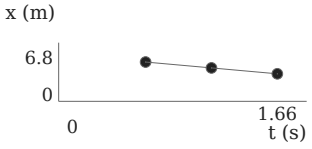
4. Photogates record the position of a cart traveling at constant velocity. t (s) -> x (m): (0.66, 6.68); (1.16, 5.68); (1.66, 4.68). With vertical error bounded by 0.12 and an exact model, what is the error bound on an inferred X?

(A) -0.19

(B) 0.31

(C) 0.06

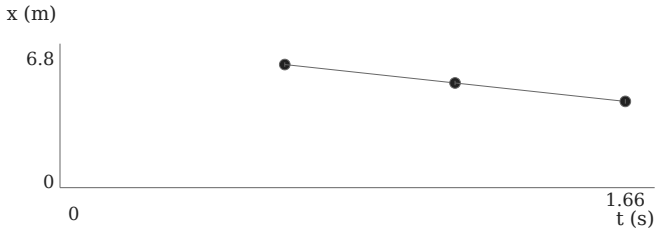
(D) 0.56



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Photogates record the position of a cart traveling at constant velocity. t (s) -> x (m): (0.66, 6.68); (1.16, 5.68); (1.66, 4.68).

a. For the track photogates setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Reverse the positive position direction only. Predict the new slope and intercept and explain physically what changed. **(3 pt)**

6. A bicycle moving right brakes uniformly. Use the model only until it stops. t (s) -> v (m/s): (0.66, 8.68); (1.16, 7.68); (1.66, 6.68).

a. Build the model for brake recorder from physics and predict both slope and intercept. **(2 pt)**

b. Double braking force at the same starting speed and mass. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A runs at +5 m/s and B at +2 m/s. Record the lead of B ahead of A as a signed separation. t (s) -> xB-xA (m): (0.66, 7.02); (1.16, 5.52); (1.66, 4.02). Build the model for runner separation from physics and predict both slope and intercept. **(3 pt)**